

Fascicule 3 : La Cryptographie avec OpenSSL

1. CHIFFREMENT SYMETRIQUE

```
(ibtihel@kali)-[~]
$ mkdir TP_CRYPTO
cd TP_CRYPTO

mkdir symetrique asymetrique hash signature certificat

(ibtihel@kali)-[~/TP_CRYPTO]
$ echo "Bonjour je suis ibtihel" > ibtihelCC.txt

(ibtihel@kali)-[~/TP_CRYPTO]
$ openssl enc -rc4 -in ibtihelCC.txt -out symetrique/fichier_rc4.enc
enter RC4 encryption password:
Verifying - enter RC4 encryption password:
*** WARNING : deprecated key derivation used.
Using -iter or -pbkdf2 would be better.

(ibtihel@kali)-[~/TP_CRYPTO]
$ openssl enc -rc4 -d -in symetrique/fichier_rc4.enc -out symetrique/fichier_rc4_dec.txt
enter RC4 decryption password:
*** WARNING : deprecated key derivation used.
Using -iter or -pbkdf2 would be better.

(ibtihel@kali)-[~/TP_CRYPTO]
$ diff ibtihelCC.txt symetrique/fichier_rc4_dec.txt

(ibtihel@kali)-[~/TP_CRYPTO]
$ openssl enc -des -in ibtihelCC.txt -out symetrique/fichier_des.enc -k 0123456789ABCDEF
```

```
(ibtihel@kali)-[~/TP_CRYPTO]
$ openssl enc -rc4 -d -in symetrique/fichier_rc4.enc -out symetrique/fichier_rc4_dec.txt
enter RC4 decryption password:
*** WARNING : deprecated key derivation used.
Using -iter or -pbkdf2 would be better.

(ibtihel@kali)-[~/TP_CRYPTO]
$ diff ibtihelCC.txt symetrique/fichier_rc4_dec.txt

(ibtihel@kali)-[~/TP_CRYPTO]
$ openssl enc -des -in ibtihelCC.txt -out symetrique/fichier_des.enc -k 0123456789ABCDEF
*** WARNING : deprecated key derivation used.
Using -iter or -pbkdf2 would be better.

(ibtihel@kali)-[~/TP_CRYPTO]
$ openssl enc -des -d -in symetrique/fichier_des.enc -out symetrique/fichier_des_dec.txt -k 0123456789ABCDEF
*** WARNING : deprecated key derivation used.
Using -iter or -pbkdf2 would be better.

(ibtihel@kali)-[~/TP_CRYPTO]
$ diff ibtihelCC.txt symetrique/fichier_des_dec.txt
diff: ibtihelCC.txt: No such file or directory

(ibtihel@kali)-[~/TP_CRYPTO]
$ diff ibtihelCC.txt symetrique/fichier_des_dec.txt
```

2. CHIFFREMENT ASYMETRIQUE

```
(ibtihel@kali)-[~/TP_CRYPT0]
$ openssl genrsa -out asymetrique/rsakey.pem 1024

(ibtihel@kali)-[~/TP_CRYPT0]
$ openssl rsa -in asymetrique/rsakey.pem -text -noout
Private-Key: (1024 bit, 2 primes)
modulus:
 00:c5:20:57:9f:ba:f9:17:e8:4d:17:8c:99:80:fe:
 30:0b:c9:77:05:ac:e5:6a:58:59:d1:c6:85:b6:64:
 bd:e0:11:db:d8:b9:a1:82:1c:1e:62:6b:ae:8d:da:
```

```
(ibtihel@kali)-[~/TP_CRYPT0]
$ openssl rsa -in asymetrique/rsakey.pem -pubout -out asymetrique/rsapubkey.pem
writing RSA key

(ibtihel@kali)-[~/TP_CRYPT0]
$ openssl enc -aes256 -in asymetrique/rsakey.pem -out asymetrique/rsakeyencaes.pem
enter AES-256-CBC encryption password:
Verifying - enter AES-256-CBC encryption password:
*** WARNING : deprecated key derivation used.
Using -iter or -pbkdf2 would be better.

(ibtihel@kali)-[~/TP_CRYPT0]
$ openssl pkeyutl -encrypt -pubin -inkey asymetrique/rsapubkey.pem -in ibtihelCC.txt -out asym
etrique/fichier_rsa.enc

(ibtihel@kali)-[~/TP_CRYPT0]
$ openssl pkeyutl -decrypt -inkey asymetrique/rsakey.pem -in asymetrique/fichier_rsa.enc -out
asymetrique/fichier_rsa_dec.txt

(ibtihel@kali)-[~/TP_CRYPT0]
$ diff fichier_nom_eleve.txt asymetrique/fichier_rsa_dec.txt
diff: fichier_nom_eleve.txt: No such file or directory

(ibtihel@kali)-[~/TP_CRYPT0]
$ diff ibtihelCC.txt asymetrique/fichier_rsa_dec.txt
```

3. SIGNATURE NUMERIQUE

```
(ibtihel@kali)-[~/TP_CRYPT0]
$ openssl rsautl -sign -in hash/fichier.sha1 -inkey asymetrique/rsakey.pem -out signature/fich
ier.sig
The command rsautl was deprecated in version 3.0. Use 'pkeyutl' instead.

(ibtihel@kali)-[~/TP_CRYPT0]
$ openssl pkeyutl -sign -in hash/fichier.sha1 -inkey asymetrique/rsakey.pem -out signature/fichier.sig

(ibtihel@kali)-[~/TP_CRYPT0]
$ ls signature/
fichier.sig
```

4. CERTIFICAT NUMERIQUE

```
(ibtihel@kali)-[~/TP_CRYPT0]
└─$ openssl req -new -x509 -days 365 -key certificat/server_cle.pem -out certificat/server_cert.crt
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
.
Country Name (2 letter code) [AU]:TN
State or Province Name (full name) [Some-State]:tunis
Locality Name (eg, city) []:tunis
Organization Name (eg, company) [Internet Widgits Pty Ltd]:esprim
Organizational Unit Name (eg, section) []:info
Common Name (e.g. server FQDN or YOUR name) []:ibtihel
Email Address []:ibtiheljenzri2@gmail.com

(ibtihel@kali)-[~/TP_CRYPT0]
└─$ ls signature/
fichier.sig

(ibtihel@kali)-[~/TP_CRYPT0]
└─$ openssl x509 -in certificat/server_cert.crt -text -noout
Certificate:
  Data:
    Version: 3 (0x2)
    Serial Number:
      28:b0:ce:f0:11:e5:b2:ae:bc:c0:7f:4a:49:b0:cc:8b:aa:6c:73:8f
    Signature Algorithm: sha256WithRSAEncryption
    Issuer: C=TN, ST=tunis, L=tunis, O=esprim, OU=info, CN=ibtihel, emailAddress=ibtiheljenzri2@gmail.com
    Validity
      Not Before: Apr 13 10:40:14 2026 GMT
      Not After : Apr 13 10:40:14 2027 GMT
    Subject: C=TN, ST=tunis, L=tunis, O=esprim, OU=info, CN=ibtihel, emailAddress=ibtiheljenzri2@gmail.com
    Subject Public Key Info:
      Public Key Algorithm: rsaEncryption
        Public-Key: (1024 bit)
        Modulus:
```

Session Actions Edit View Help

ibtihel@kali: ~

ibtihel@kali: ~

ibtihel@kali: ~/TP_CRYPTO

```
X509v3 extensions:
  X509v3 Subject Key Identifier:
    F4:41:00:5F:3E:0C:AF:AA:9B:97:BE:23:C2:B7:9F:92:44:19:D4:66
  X509v3 Authority Key Identifier:
    F4:41:00:5F:3E:0C:AF:AA:9B:97:BE:23:C2:B7:9F:92:44:19:D4:66
  X509v3 Basic Constraints: critical
    CA:TRUE
Signature Algorithm: sha256WithRSAEncryption
Signature Value:
  86:81:a0:e3:15:af:97:1e:ca:c7:a7:e7:a2:80:d2:71:78:46:
  55:0b:36:c9:28:c9:e0:03:4a:1e:16:ab:2d:8a:be:a9:0d:6d:
  18:1f:2f:87:6f:be:14:7e:63:df:57:a2:ad:13:69:c5:ab:32:
  14:67:05:03:a2:f9:0a:e8:a0:9e:2d:6e:4d:c8:f2:58:0b:e9:
  9e:ed:4a:31:24:d5:cd:02:ae:1f:51:11:0e:d4:1c:bb:07:fa:
  1f:80:b3:14:64:f5:14:0a:ef:3c:4b:04:34:2b:de:2b:7e:24:
  3a:12:72:c2:50:39:94:e5:93:80:bd:f6:36:fa:87:58:c1:46:
  83:1f
```

(ibtihel@kali)~/TP_CRYPTO

\$ openssl genrsa -out certificat/cakey.pem 2048

(ibtihel@kali)~/TP_CRYPTO

\$ openssl req -new -x509 -days 730 -key certificat/cakey.pem -out certificat/ca.crt

You are about to be asked to enter information that will be incorporated into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.

Country Name (2 letter code) [AU]:TN
State or Province Name (full name) [Some-State]:tunis
Locality Name (eg, city) []:tunis
Organization Name (eg, company) [Internet Widgits Pty Ltd]:esprim
Organizational Unit Name (eg, section) []:ibtihel
Common Name (e.g. server FQDN or YOUR name) []:ibtihel^C

(ibtihel@kali)~/TP_CRYPTO

\$ ls
asymetrique certificat hash ibtihelCC.txt signature symetrique

(ibtihel@kali)~/TP_CRYPTO

\$ ls -R

.:
asymetrique certificat hash ibtihelCC.txt signature symetrique

./asymetrique:
fichier_rsa_dec.txt fichier_rsa.enc rsakeyencaes.pem rsakey.pem rsapubkey.pem

./certificat:
cakey.pem server_cert.crt server_cle.pem server.csr

./hash:
fichier.md5 fichier.sha1

./signature:
fichier.sig

./symetrique:
fichier_des_dec.txt fichier_des.enc fichier_rc4_dec.txt fichier_rc4.enc

(ibtihel@kali)~/TP_CRYPTO

\$